



AI CAREERS FOR WOMEN LINKEDIN FELLOWSHIP

Sales & Business Data Cleaning and Visualization

Project Domain

Excel / Power BI

A Collaborative Initiative

equipping students with practical skills in AI, Data Analytics, and Business Intelligence

by:



Faculty Name

Sagar Chetan Mehta



Student Email

nadkaramruta42@gmail.com



Guide

Umamaheswari

Institution:

Smt. Maniben M.P. Shah Women's College of Arts & Commerce

Year - 2026

3. Problem Statement

The e-commerce industry generates large volumes of sales data every day. However, this raw data often contains inconsistencies, missing values, duplicate records, and unstructured formats, making it difficult to extract meaningful insights.

In this project, a sales dataset from **Flipkart** will be analyzed to clean and organize the data, identify sales patterns, and evaluate product performance, regional demand, and profitability. The goal is to transform raw sales data into meaningful insights using Excel and AI-assisted analysis.

This will help businesses understand customer purchasing behavior, optimize product strategies, and support data-driven decision making.

4. Objectives

List the specific goals your project aims to accomplish. Objectives should be clear and measurable.

- To explore and understand the structure of the Flipkart sales dataset including product category, sales amount, profit, region, and customer segment.
- To clean and preprocess the dataset by removing duplicates, handling missing values, and correcting data formats.
- To analyze sales performance based on **product category, region, and customer segment**.
- To identify **top-selling products, highest revenue categories, and profitable regions**.
- To create an **interactive dashboard in Excel** showing key sales KPIs.
- To use **AI tools such as ChatGPT** to generate insights and summaries from the analyzed data.

5. Methodology / Implementation Plan

Provide a simple overview of how the project will be carried out. The following five-phase workflow applies to all projects under this fellowship program.

#	Phase	Description
1	Data Collection & Exploration	The project uses a Flipkart sales dataset containing information such as product name, category, price, quantity sold, order date, customer rating, region, and profit. The dataset will be explored to understand field names, data types, and data distribution.

2	Data Cleaning & Preprocessing	The dataset will be cleaned by removing duplicate records, handling missing values, correcting date and numerical formats, and standardizing text fields such as product categories and regions.
3	Analysis & AI Integration	Excel formulas, pivot tables measures will be used to analyze key metrics such as total sales, profit, quantity sold, and average customer ratings. AI tools will be used to generate summaries and insights from the analysis.
4	Visualization / Output	An interactive dashboard will be created showing: • Total Sales and Profit KPIs • Category-wise Sales • Region-wise Sales Performance • Monthly Sales Trends • Top Performing Products
5	Documentation & Submission	The final report will document the complete workflow including dataset cleaning, analysis, dashboard creation, and AI-assisted insights.

6. Resources and Timeline

A. Hardware & Software Requirements

- Laptop or PC — Intel Core i3 or higher, minimum 4 GB RAM (8 GB recommended)
- Operating System: Windows 10 / 11 or macOS
- Microsoft Excel (Microsoft 365 / Excel 2019 or later)
- Power BI Desktop — latest version (free download from Microsoft) — where applicable
- Internet connection — required for Copilot and AI Chatbot access
- Web browser — Google Chrome or Microsoft Edge (latest version)

B. Dataset

- **Dataset Availability**
The dataset used for this project is already available in CSV format and contains sales-related information such as product name, category, price, quantity sold, discount, region, profit, and customer segment. This dataset represents e-commerce sales transactions that can be analyzed to understand business performance.
- **Dataset Source**
The dataset is publicly available from **Kaggle**, a popular platform that provides open

datasets for data analysis and machine learning projects. It is a sample sales dataset inspired by the operations of **Flipkart**, used for educational and analytical purposes.

- **Data Usage in the Project**

The dataset will be imported into Microsoft Excel where it will be cleaned, processed, and analyzed. Techniques such as removing duplicates, handling missing values, and standardizing formats will be applied before performing analysis.

- **Purpose of Using the Dataset**

The dataset will help analyze key business metrics such as total sales, profit, regional performance, product demand, and customer segments. These insights will support data-driven decision making and demonstrate practical data analytics skills.

C. Project Timeline

A simple table showing planned tasks and expected completion dates.

Phase	Task	Tentative Completion Date
1	Dataset collection, exploration, and understanding	<i>Week 1</i>
2	Data cleaning, preprocessing, and validation / Prompt design	<i>Week 2</i>
3	Analysis, AI tool integration (Copilot / Chatbot), and insight generation	<i>Week 2 – 3</i>
4	Dashboard / output build, testing, and review	<i>Week 3</i>
5	Report writing, GitHub upload, and video recording and submission	<i>Week 4</i>

7. Expected Outcomes / Conclusion

This section summarizes what the team expects to deliver by the end of the project. Outcomes should be realistic and directly tied to the objectives stated above.

- A **clean and structured Flipkart sales dataset** with all preprocessing steps documented.
- A **data analysis report identifying sales trends, product performance, and regional demand**.
- An **interactive dashboard** displaying key sales KPIs and visual insights..
- Demonstrated use of **AI tools to generate analytical summaries and insights** from the dataset.
- **Project documentation** including the dataset, dashboard, and report.

This project analyzes an e-commerce sales dataset inspired by **Flipkart** to demonstrate the process of data cleaning, analysis, and visualization. By organizing and analyzing the dataset using tools like Excel, meaningful insights about sales performance, product demand, regional trends, and profitability can be identified.

The project shows how structured data analysis and AI tools can help transform raw data into valuable business insights, supporting better decision-making and improving overall business performance.

8. References

List all references, tutorials, and documentation used in the project. Students must cite any material they referred to, including research papers, GitHub repositories, YouTube tutorials, blogs, and official documentation.

- Edunet Foundation. AICW Fellowship Program Guidelines. 2026. Available at: <https://www.edunetfoundation.org>
- Microsoft Corporation. Microsoft Excel Documentation. Available at: <https://support.microsoft.com/en-us/excel>
- Microsoft Corporation. Power BI Documentation. Available at: <https://learn.microsoft.com/en-us/power-bi/>
- Kaggle Inc. Open Datasets. Available at: <https://www.kaggle.com/datasets>